

Zhongxin Hu

PhD Candidate at Eindhoven University of Technology

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Research Profile

My research focuses on sustainable operations management. Using analytical modeling, game theory, contract theory, and optimization under uncertainty, I study circular business models such as remanufacturing and refurbishment and examine how firms can make economically, environmentally, and socially beneficial decisions.

My broader interests include pricing and equilibrium analysis in queueing systems, behavioral operations, and sequential games.

Education

Eindhoven University of Technology (TU/e)

Eindhoven, NL

PhD Candidate in Operations Management

July 2023 – June 2027

- Supervisors: Zümbül Atan and Christina Imdahl
- Group: Operations, Planning, Accounting & Control (OPAC)

Beijing University of Posts and Telecommunications (BUPT)

Beijing, CN

M.Sc. in Mathematics

Sep. 2020 – Jun. 2023

- Supervisor: Yongjiang Guo
- Thesis: *Information Disclosure and Reference Dependence in the Cooperative Newsvendor Problem*

Nanjing University of Posts and Telecommunications

Nanjing, CN

B.Sc. in Applied Statistics

Sep. 2015 – Jun. 2019

Working Papers

- **Profit Meets Purpose: Refurbish-to-Lease Model for Medical Essentials**

Zhongxin Hu, Christina Imdahl, Yeqing Zhou (RSM), Zümbül Atan.

Conference presentations:

2026 OPAC Conference (*Best Presentation Award*), Heeze, NL; Circularity and Sustainability Workshop (Poster), Utrecht, NL.

2025 INFORMS Annual Meeting, Atlanta, USA; Beta Symposium, Soesterberg, NL.

- **Strategic Selection of Remanufacturing Business Models: A Consumer Perception Perspective**

Zhongxin Hu, Christina Imdahl, Zümbül Atan.

Status: Under review.

Conference presentations:

2026 OPAC Conference, Heeze, NL.

2025 International Symposium on Inventories, Budapest, HU.

Publications

- [1] Cao, J., Guo, Y., **Hu, Z.** *The Effect of Loss Preference on Queueing with Information Disclosure Policy.* Methodology and Computing in Applied Probability, 25: 75, 2023.
- [2] **Hu, Z.**, Guo, Y. *Cooperating Strategy of Newsvendor with Advertisers Under Partial Quality Disclosure.* Journal of Systems Science and Mathematical Sciences, 43(3): 667–682, 2023.

Research & Project Experience

- LINear to Circular Transition (LINCIT)

July 2023 – June 2026

 - Analyzed operational decisions in circular business models, including remanufacturing and refurbishment.
 - Presented annual research progress to consortium partners.
- Post-Graduate Innovation and Entrepreneurship Project

Apr. 2022 – Nov. 2022

 - Built analytical models combining sequential game theory, prospect theory, and contract theory to study computational-resource allocation in blockchain systems.
- China Contemporary Undergraduate Mathematical Contest in Modeling

Sep. 2017

 - Developed a joint pricing and allocation model for crowdsourced task platforms and used numerical experiments to derive allocation policies.

Teaching Experience

- Teaching Assistant

 - BSc courses:** Fundamentals of Operations Management (TU/e, Q3 2024/25 & 2025/26); Probability Theory and Stochastic Processes (BUPT, Spring 2021); Advanced Mathematics (BUPT, Fall 2020).
 - MSc course:** Reinforcement Learning: Theory and Applications (BUPT, Summer 2021).
- Bachelor’s Thesis Supervision

 - Supervised 4 Bachelor End Projects at TU/e (2024/25–2026/27).

Awards & Honors

- Best Presentation Award**, OPAC Conference, Heeze, NL (2026).
- Outstanding Graduate Student Award**, BUPT (2023).
- Provincial First Prize**, China Contemporary Undergraduate Mathematical Contest in Modeling, Jiangsu Province (2017).
- Merit Undergraduate Student Award**, NJUPT (2016).

Academic Service & Activities

- Volunteer Mentor, Lumiere Research Inclusion Foundation, Lumiere Education

2026
- Session Chair, INFORMS Annual Meeting, Atlanta, USA

2025
- Reviewer, *Journal of Systems Science and Mathematical Sciences*

2023
- Fellow, SheShapes Academia 2022, Global Shapers Community Beijing Hub

2022